P400

Multi-Frequency 900 & 400 MHz Wireless Modem

For the ultimate solution in design flexibility and cost/size sensitive applications, consider the Pico Series P400 radio platform. Offering software selectable ISM 900 MHz & Licensed 400MHz modes, in a single module, the P400 provides an economical solution with the design flexibility, features, and performance never seen before!

Weighs only 5 grams!



1.3" X 1.05" X 0.13"

- ***400 MHz Licensed***
- ***900 MHz ISM***
- ***Up to 2 Watts***
- ***Extended Temperature***
- ***Dual Serial Ports***
- ***Excellent Sensitivity***

Features

- Supports up to 345 kbps (@ 900MHz)
- Software Selectable 400 MHz & 900 MHz Bands.
- Quad Filter Stage provides Extreme Noise & Interference Rejection
- Point-to-Point, Point-to-Multipoint, Store & Forward Repeater
- Industrial Temperature (-55°C to +85°C)
- Maximum allowable transit power (Adjustable)
- Low Power consumption in Sleep and Sniff modes
- 32 bits of CRC, selectable Forward Error Correction with retransmission
- Separate diagnostics port—transparent remote diagnosis and online network control
- Extremely Small Footprint (26.5 x 33 x 3.5mm | 1.3 x 1.05 x 0.15")
- Compatible with some GPS Radio Transceivers
- Compatible with Microhard 920F



Enclosed Option

P400-SWP

Specifications

Spreading Method / Modulation Scheme	Frequency Hopping, GMSK, 2GFSK, 4GFSK, QPSK
Forward Error Correction	Hamming, BCH, Golay, Reed-Solomon, Viterbi
Error Detection	32 bits of CRC, ARQ
Encryption	Optional (see –AES option)
Range	60 miles (100 km)
Serial Interface	3.3V CMOS RS232/485 (Selectable)
Serial Baud Rate	300 bps to 230 kbps
Operating Modes	Point-to-Point, Point-to-Multipoint, Store & Forward Repeater, Peer-to-Peer
Signals Interface	RSSI LEDs, Tx/Rx LEDs, Reset, Config, Wake-up, RSmode, 4 Digital Inputs/Outputs, 1 Analog Input, 1 Analog Output
Remote Diagnostics	Battery Voltage, Temperature, RSSI, Packet Statistics
Rejection	Adjacent Channel @ 400 MHz: 60 dB Alternate Channel @ 400 MHz: 70 dB Adjacent Channel @ 900 MHz: 57 dB Alternate Channel @ 900 MHz: 65 dB
Core Voltage	
OEM	3.3VDC is required for 1W 3.6VDC is required for 2W
Enclosed	9-30 VDC
SWP	7-30 VDC
Power Consumption (3.3V)	Sleep: < 1mA (Future) Idle: 20mA Rx: 45mA to 98mA Tx Peak: 2A
Connectors:	
OEM	Antenna: UFL Data: 80 Pin SMT
Enclosed	Antenna: RP-SMA Female Bulkhead Data: DB9-F
SWP	Antenna: MMCX Power, Data: 10-Pin (GHR-10V-S)
Environmental	-55°C to +85°C 5-95% humidity, non-condensing
Weight	
OEM	~ 5 grams
Enclosed	~ 120 grams
SWP	~ 37 grams
Dimensions	
OEM	~ 26.5 x 33 x 3.5 mm
Enclosed	~ 46 x 66 x 25 mm
SWP	~ 36 x 48 x 15 mm
Approvals	FCC Part 15.247 IC RSS210 FCC Part 15.90 IC RSS119 RoHS Compliant

Rate (kbps)	Power	Sensitivity (dBm)	Bandwidth (kHz)	Regulatory
Frequency 410 to 480 MHz (Licensed Band)				
3.6	2W	-118	6.25	FCC / IC
4.8	2W	-117	12.5	FCC / IC
9.6	2W	-115	12.5	FCC / IC
19.2	2W	-114	25	IC
Frequency 410 to 480 MHz (Frequency Hopping)				
56	2W**	-113	60	None*
115.2	2W**	-109	150	None*
172.8	2W**	-108	180	None*
230.4	2W**	-106	230	None*
276.4	2W**	-105	230	None*
345	2W**	-103	400	None*
Frequency 902 to 928 MHz (Frequency Hopping)				
19.2	1W	-116	25	FCC / IC
56	1W	-113	60	FCC / IC
115.2	1W	-109	150	FCC / IC
172.8	1W	-108	180	FCC / IC
230.4	1W	-106	230	FCC / IC
276.4	1W	-105	230	FCC / IC
345	1W	-103	400	FCC / IC
19.2	2W**	-115	25	None*
56	2W**	-110	60	None*
115.2	2W**	-109	150	None*
172.8	2W**	-108	180	None*
230.4	2W**	-106	230	None*
276.4	2W**	-105	230	None*
345	2W**	-103	400	None*

Order Options

P400	Base Model (1W 900MHz & 2W 400MHz Licensed Operation).*
-AES	128-bit AES Encryption.**
-C2S	2W 900MHz, 2W 400MHz Frequency Hopping, 2W 400MHz Licensed & 128-bit AES.**
-C1S	1W 900MHz, 1W 400MHz Frequency Hopping, 2W 400MHz Licensed & 128-bit AES.**
-ENC	Enclosed Model
-SWP	SWP Series

***Standard modems are shipped with 400MHz Licensed band operation up to 2W and 900MHz ISM FHSS operation 1W with no AES encryption. No other operation is allowed.**

Operating outside this requires compliance with applicable radio regulatory bodies and Canadian export laws. Extra cost/activation/proof of Regulatory Compliance is required.

****AES encryption, 2W frequency hopping operation requires an Export Permit.**